Bryan Tran

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Security Clearance: Top Secret (Active)

Education

Georgia Institute of Technology **Aug. 2026 – Present**
Master of Science in Computer Science *Atlanta, GA*

Algoverse AI Safety Research Fellowship **July 2026 – Aug. 2026**
AI Safety Research Fellow *Remote*

- Researching how vision-language models (VLMs) lose information when interpreting images, and tracing where that lost information propagates within the model.

Louisiana State University **Aug. 2019 – May 2023**
Bachelor of Science in Computer Science *Baton Rouge, LA*

Experience

IBM **June 2023 – Present**
Application Developer *Baton Rouge, LA*

- Utilized **Amazon Bedrock (Claude)** to cross-reference legacy **XGen** logic with **DMN/DRL** rules, iteratively engineering prompts and debugging AI-generated **Java** tests, then led KT sessions to standardize the resulting validation framework across the engineering team.
- Reverse-engineered legacy **VBA** business logic and multi-condition eligibility rules embedded in a **Microsoft Access** application, translating them into a layered **Angular** + **Spring Boot** + **Oracle** architecture while preserving existing behavior across the full stack.
- Designed a **REST API** preview/validation layer (**Spring Boot**, **JDBC Template**) surfacing eligible-record and key-violation counts ahead of batch database operations, replacing an opaque legacy Access confirmation flow with an explicit, inspectable API contract.
- Modernized a DFAS enterprise payroll application by building the corresponding **Angular** and **TypeScript** frontend and **REST API** integration, improving data retrieval speed over the legacy Access workflow.
- Developed backend services in **Java** and authored custom unit tests using **Mockito**, achieving **90% code coverage** on JaCoCo reports.
- Debugged full-stack integration issues across the request path, diagnosing **HTTP 401** authentication failures, **HTTP 200** responses that failed JSON parsing, and **TypeScript**/API contract mismatches between **Spring Boot** DTOs and **Angular** interfaces.

Louisiana State University, Information Technology Services **May 2022 – May 2023**
Student Web Developer *Baton Rouge, LA*

- Diagnosed and resolved production application failures for LSU's enterprise application (**GROK**) through log analysis, system walkthroughs, and proactive error tracing.
- Hardened application security by migrating legacy **Java** applications to **Spring Boot**, updating **Maven** builds, and reconfiguring **CSRF tokens** to resolve deployment blockers.
- Optimized accessibility and responsive design layouts across university applications using **HTML**, **CSS**, and modern web dependencies.

Projects

QuantDrill | *React, TypeScript, Vite, Tailwind, Vitest* **quant.bhtran.com**

- Built a full-stack quant-aptitude trainer replicating prop-trading firm assessment formats (e.g. Optiver) across **7 timed drills** covering probability, sequences, and order-book arbitrage.
- Engineered pure-function question generators with **exact-solution grading**, validated via **Vitest** unit tests, and deployed with **zero backend dependency** and automated **CI/CD**.

Sift | *Next.js, PostgreSQL, Drizzle ORM, GitHub Actions* **sift.bhtran.com**

- Developed a self-hosted job aggregation platform ingesting **∼9,000 postings per run** from Greenhouse/Workday ATS APIs across **45 curated companies**, deduplicating to **∼7,100 unique roles**.
- Designed **idempotent diffing** and **fan-out collapse** logic to track posting lifecycle over time, automated via a **30-minute GitHub Actions** ingest schedule.

Technical Skills

Languages: Python, TypeScript, Java, C#, JavaScript, SQL, PHP, HTML/CSS
Frameworks & Libraries: Next.js, React, Angular, Streamlit, Spring Boot, FastAPI, Node.js, NumPy, Pandas, PyTorch, Hugging Face, Vite, Tailwind CSS, Drizzle ORM
Cloud & Databases: AWS, Azure, Docker, PostgreSQL, MySQL, Oracle, Vercel, Supabase
Architecture & Tools: REST APIs, Temporal, Git/GitLab, GitHub Actions, Postman, JUnit, Mockito, Vitest, Maven, Jira